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Paper Code: TBT-604a

END SEMESTER EXAMINATION, JUNE 2025

B. Tech Biotechnology, Semester VI

Biosimilar Technology

Time: 3 Hr

MM: 100

Note:

- (i) Answer **all the questions** by choosing *any two of the sub questions*.
(ii) Each question carries 10 marks

Q1	(10×2 = 20 marks)	
(a)	Elaborate the regulatory guidelines of India for biosimilar development.	CO1
(b)	Explain the market potential of biosimilar development in India.	
(c)	Describe stepwise the prerequisites of biosimilar and conditions for its approval.	
Q2	(10×2 = 20 marks)	
(a)	Write a note on: antibody biologics and explain at least five examples of it	CO2
(b)	What is stem cell therapy? Give examples of FDA approved cell-based therapies and their indications.	
(c)	"Biosimilars reduces the cost of biologics many fold" Explain using any example.	
Q3	(10×2 = 20 marks)	
(a)	Describe the techniques used for adsorption of protein and peptides on different surfaces.	CO3
(b)	Explain the N-terminal methods of protein sequencing?	
(c)	What are the factors that impact protein stability and how the stability can be enhanced.	
Q4	(10×2 = 20 marks)	
(a)	Discuss the experimental design and statistical issues involved in bioequivalence studies.	CO4
(b)	What are the requirements for a biosimilar to be designated inter-changeable by the FDA?	
(c)	Elucidate the role and basis of PURPLE BOOK.	
Q5	(10×2 = 20 marks)	
(a)	Discuss a case study for AstraZeneca's long-acting monoclonal antibody combination, AZD7442 (Evusheld) for COVID-19 preexposure prophylaxis in adults and children at least 12 years old.	CO5
(b)	Highlight the key biosimilar research mandate of any two biotech companies.	
(c)	Discuss a case study for Interferon alpha-2b biosimilar developed by Zydus Cadila.	

